

CSE 4412: Introduction to Quantum Computing

Homework 1 — Chapter 2 Exercises

Exercise 2.2

Statement. Let $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. Compute $\langle +|+\rangle$, $\langle +|-\rangle$, and $\langle -|-\rangle$. What can you say about these two vectors? You may use the fact that $|0\rangle$ and $|1\rangle$ are orthonormal.

Hint. Expand each bracket using bilinearity of the inner product (distribute over the $\pm$ sums), then substitute $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = \langle 1|0\rangle = 0$. Once you have all three numbers, think about what “normalized” and “orthogonal” mean (Definitions in Section 2.1.2) and what it implies when two normalized, mutually orthogonal vectors span the whole space.

Your answer:

Expand each inner product using bilinearity, then substitute $\langle 0|0\rangle = \langle 1|1\rangle = 1$ and $\langle 0|1\rangle = \langle 1|0\rangle = 0$.

$$\langle +|+\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (\langle 0|0\rangle + \langle 0|1\rangle + \langle 1|0\rangle + \langle 1|1\rangle) = \frac{1}{2}(1 + 0 + 0 + 1) = 1$$

$$\langle +|-\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (\langle 0|0\rangle - \langle 0|1\rangle + \langle 1|0\rangle - \langle 1|1\rangle) = \frac{1}{2}(1 - 0 + 0 - 1) = 0$$

$$\langle -|-\rangle = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} (\langle 0|0\rangle - \langle 0|1\rangle - \langle 1|0\rangle + \langle 1|1\rangle) = \frac{1}{2}(1 - 0 - 0 + 1) = 1$$

Since $\langle +|+\rangle = \langle -|-\rangle = 1$, both $|+\rangle$ and $|-\rangle$ are normalized (unit vectors), and since $\langle +|-\rangle = 0$, they are orthogonal to each other. Two normalized, mutually orthogonal vectors in a 2-dimensional space form an orthonormal basis, so $\{|+\rangle, |-\rangle\}$ is an alternative orthonormal basis for the qubit state space (alongside $\{|0\rangle, |1\rangle\}$).

Exercise 2.4

Statement. Consider the following three questions:

1. Consider the $|+\rangle$ state. What is the probability of observing $|1\rangle$ after a measurement?
2. Consider $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$. What is the probability of observing $|1\rangle$ after measuring that state?
3. Consider the state $|1\rangle$. What is the probability of observing $|1\rangle$ after measurement?

Hint. For a state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the rule from the chapter is $\Pr(|1\rangle) = |\beta|^2$, where β is whatever is multiplying $|1\rangle$ once the state is written in the $\{|0\rangle, |1\rangle\}$ basis. Identify β in each of the three given states (watch the sign in $|-\rangle$ — remember you square the *magnitude*).

Your answer:

For a state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $\Pr(|1\rangle) = |\beta|^2$.

1. $|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$, so $\beta = \frac{1}{\sqrt{2}}$ and

$$\Pr(|1\rangle) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$

2. $|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$, so $\beta = -\frac{1}{\sqrt{2}}$. Squaring the *magnitude* discards the sign, so

$$\Pr(|1\rangle) = \left| -\frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2},$$

the same probability as for $|+\rangle$, even though the two states are orthogonal (Exercise 2.2) — the relative phase (sign) between α and β does not affect the measurement probabilities in the computational basis.

3. $|1\rangle = 0 \cdot |0\rangle + 1 \cdot |1\rangle$, so $\beta = 1$ and

$$\Pr(|1\rangle) = |1|^2 = 1,$$

i.e. measuring $|1\rangle$ always yields “1” with certainty, as expected since the state already is a computational basis state.

Exercise 2.5

Statement. Imagine you are given a qubit $|\psi\rangle$ but not told anything about it (i.e., you do not know its probability amplitudes). If you perform a measurement on the state and the outcome is “0,” what does this tell you about the state? Can you learn the original probability amplitudes? What if you are given as many copies of $|\psi\rangle$ as you want — what can be said then?

Hint. Separate the question into two parts. (a) Single measurement: think about *state collapse* — what do you know about the state right after the measurement, versus what you knew about it beforehand? Can one classical bit of outcome data pin down two continuous amplitudes α, β ? (b) Many copies: if you measured each copy and tallied the fraction of “0” vs. “1” outcomes, what would that fraction converge to as the number of copies grows (law of large numbers)? Does that let you recover α and β exactly, or only something about their magnitudes? Also recall the no-cloning theorem (mentioned in the text, Section 2.5.4) — why can’t you manufacture the extra copies yourself?

Your answer:

Write $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with unknown $\alpha, \beta \in \mathbb{C}$, $|\alpha|^2 + |\beta|^2 = 1$.

(a) Single measurement. Before the measurement, the state carries two continuous, real degrees of freedom (e.g. $|\alpha|$ and the relative phase, since one global degree of freedom is fixed by normalization and another which the global phase is physically unobservable). A measurement in the computational basis produces only *one classical bit* of outcome data (“0” or “1”), which cannot possibly encode two continuous parameters – so a single outcome can never pin down α, β exactly. Concretely, observing “0” tells us only that $\alpha \neq 0$ (this outcome had nonzero probability $|\alpha|^2$), and by *state collapse* the qubit is now *exactly* $|0\rangle$, regardless of what α, β were beforehand. So the single measurement destroys the very information we wanted to learn: post-measurement the state is deterministically $|0\rangle$, and the original amplitudes are unrecoverable, whether they were, say, $(\alpha, \beta) = (1, 0)$ or $(\alpha, \beta) = (\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$ or anything else with $\alpha \neq 0$ – all of these are consistent with having observed “0.”

(b) Many copies. If instead we are handed n independently prepared copies of the same unknown $|\psi\rangle$ and measure each one (in the computational basis), the fraction of “0” outcomes is a random variable whose expectation is $|\alpha|^2$. By the law of large numbers, as $n \rightarrow \infty$ this empirical fraction converges to $|\alpha|^2$ exactly (and the fraction of “1” outcomes converges to $|\beta|^2 = 1 - |\alpha|^2$). So with enough copies we can estimate the *magnitudes* $|\alpha|$ and $|\beta|$ to arbitrary precision. However, this still does not recover α and β exactly: measuring only in the computational basis is blind to the *relative phase* between α and β (e.g. $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$ and $\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |-\rangle$ give identical statistics in the $\{|0\rangle, |1\rangle\}$ basis – see Exercise 2.4 parts 1–2 – yet are orthogonal states). To also determine the relative phase one needs additional copies measured in other bases (e.g. $\{|+\rangle, |-\rangle\}$); doing so for a tomographically complete set of bases lets one reconstruct α, β up to the single remaining, physically unobservable global phase (i.e. $|\psi\rangle$ and $e^{i\theta}|\psi\rangle$ can never be distinguished by any measurement).

Finally, note why the problem insists the copies must be *given* to us rather than something we produce ourselves: the **no-cloning theorem** (Section 2.5.4) says there is no physical process that takes one copy of an arbitrary, unknown state $|\psi\rangle$ and outputs two copies of it. So we cannot bypass part (a)’s limitation by measuring our single qubit, then cloning it before measuring again – copies of an unknown state can only come from someone (or something) re-preparing the same state independently each time, not from duplicating the one qubit we hold.

Exercise 2.6

Statement. Let $p \in [0, 1]$ be a real number and consider the state

$$|\psi\rangle = \sqrt{p} |0\rangle + \sqrt{1-p} |1\rangle.$$

Answer the following: (1) probability of observing $|0\rangle$ for a given p ; (2) probability of observing $|-\rangle$ for a given p ; (3)–(4) graph both probability distributions as functions of p .

Hint. For (1), same rule as Exercise 2.4, just applied to $|0\rangle$ instead of $|1\rangle$. For (2), you can’t just read off a coefficient because $|-\rangle$ isn’t one of the two basis vectors $|\psi\rangle$ is written in terms of — instead use the inner-product rule $\Pr(|-\rangle) = |\langle -|\psi\rangle|^2$ from Section 2.3.1 (Equation 2.36), and expand $\langle -|\psi\rangle$ using linearity and the known values of $\langle 0|0\rangle, \langle 0|1\rangle, \langle 1|0\rangle, \langle 1|1\rangle$. For (3)–(4), once you have closed-form expressions in terms of p , you can plot them with `pgfplots` (`\usepackage{pgfplots}`) or by hand on $[0, 1]$ — try a few sample points ($p = 0, \frac{1}{2}, 1$) first to sanity-check the shape before you commit to plotting code.

Your answer:

(1) Probability of observing $|0\rangle$. Writing $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $\alpha = \sqrt{p}$, the rule from Exercise 2.4 (applied to $|0\rangle$ instead of $|1\rangle$) gives

$$\Pr(|0\rangle) = |\alpha|^2 = |\sqrt{p}|^2 = p.$$

(Sanity check: $p = 0 \Rightarrow |\psi\rangle = |1\rangle \Rightarrow \Pr(|0\rangle) = 0$; $p = 1 \Rightarrow |\psi\rangle = |0\rangle \Rightarrow \Pr(|0\rangle) = 1$. Both match $\Pr(|0\rangle) = p$.)

(2) Probability of observing $|-\rangle$. Since $|-\rangle$ is not one of the two basis vectors $|\psi\rangle$ is expanded in, use $\Pr(|-\rangle) = |\langle -|\psi\rangle|^2$. Expand $\langle -| = \frac{1}{\sqrt{2}}(\langle 0| - \langle 1|)$ against $|\psi\rangle = \sqrt{p}|0\rangle + \sqrt{1-p}|1\rangle$:

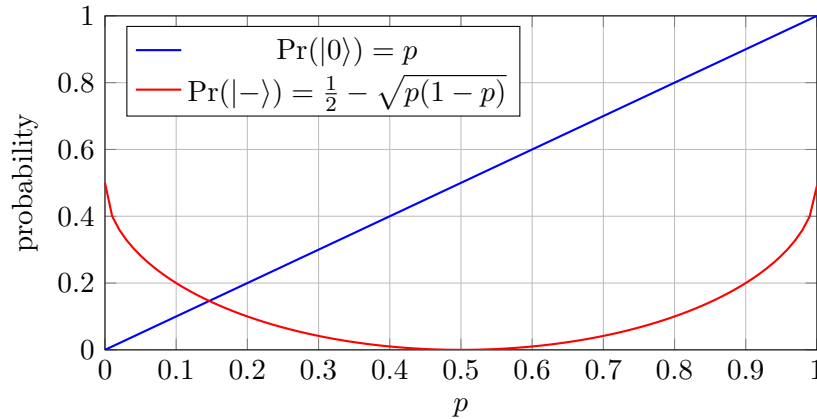
$$\langle -|\psi\rangle = \frac{1}{\sqrt{2}}(\sqrt{p}\langle 0|0\rangle + \sqrt{1-p}\langle 0|1\rangle - \sqrt{p}\langle 1|0\rangle - \sqrt{1-p}\langle 1|1\rangle) = \frac{1}{\sqrt{2}}(\sqrt{p} - \sqrt{1-p}).$$

Squaring the magnitude,

$$\Pr(|-\rangle) = |\langle -|\psi\rangle|^2 = \frac{1}{2}(\sqrt{p} - \sqrt{1-p})^2 = \frac{1}{2}(p + (1-p) - 2\sqrt{p(1-p)}) = \frac{1}{2} - \sqrt{p(1-p)}.$$

(Sanity check: $p = \frac{1}{2} \Rightarrow |\psi\rangle = |+\rangle$, and $\Pr(|-\rangle) = \frac{1}{2} - \frac{1}{2} = 0$, correct since $|+\rangle \perp |-\rangle$. At $p = 0$ or $p = 1$, $|\psi\rangle \in \{|0\rangle, |1\rangle\}$ and $\Pr(|-\rangle) = \frac{1}{2} - 0 = \frac{1}{2}$, correct since $|0\rangle, |1\rangle$ each have overlap $\frac{1}{\sqrt{2}}$ with $|-\rangle$.)

(3)–(4) Plots. Both are functions of $p \in [0, 1]$: $\Pr(|0\rangle) = p$ is a straight line from $(0, 0)$ to $(1, 1)$; $\Pr(|-\rangle) = \frac{1}{2} - \sqrt{p(1-p)}$ is symmetric about $p = \frac{1}{2}$, equal to 0 at $p = \frac{1}{2}$ (its minimum) and rising to $\frac{1}{2}$ at both endpoints $p = 0, 1$ (its maximum) – an upside-down “dome” turned into a “valley”.



Exercise 2.8

Statement. Let $|\psi\rangle$ be some d -dimensional state and let $\mathcal{B} = \{|b_0\rangle, \dots, |b_{d-1}\rangle\}$ be any orthonormal basis. Show that $|\psi\rangle$ may be written as

$$|\psi\rangle = \sum_{j=0}^{d-1} \langle b_j|\psi\rangle |b_j\rangle.$$

Hint. This is essentially asking you to re-derive the *completeness relation* argument from Section 2.1.3 (Equation 2.22-2.23). Start from the fact that $\sum_j |b_j\rangle \langle b_j| = I$ for any orthonormal basis, apply both sides to $|\psi\rangle$, and note that $|b_j\rangle \langle b_j| |\psi\rangle$ can be regrouped as a scalar $\langle b_j|\psi\rangle$ times the vector $|b_j\rangle$.

Your answer:

Since $\mathcal{B} = \{|b_0\rangle, \dots, |b_{d-1}\rangle\}$ is a basis for the d -dimensional space, every vector in that space – including $|\psi\rangle$ – can be written as *some* linear combination of the $|b_j\rangle$:

$$|\psi\rangle = \sum_{j=0}^{d-1} c_j |b_j\rangle, \quad \text{for some (as yet unknown) scalars } c_j \in \mathbb{C}.$$

To identify the coefficients c_j , take the inner product of both sides with an arbitrary basis bra $\langle b_k|$:

$$\langle b_k|\psi\rangle = \sum_{j=0}^{d-1} c_j \langle b_k|b_j\rangle.$$

Because $\mathcal{B}$ is *orthonormal*, $\langle b_k|b_j\rangle = \delta_{kj}$ (i.e. 1 if $k = j$, 0 otherwise), so every term in the sum vanishes except $j = k$:

$$\langle b_k|\psi\rangle = c_k.$$

This holds for every $k = 0, \dots, d-1$, so the unknown coefficients are exactly $c_j = \langle b_j|\psi\rangle$. Substituting back into the original expansion gives

$$|\psi\rangle = \sum_{j=0}^{d-1} \langle b_j|\psi\rangle |b_j\rangle,$$

as required. Equivalently, this is the statement that $\sum_j |b_j\rangle \langle b_j| = I$ (the *completeness relation*): applying this operator identity to $|\psi\rangle$ gives $|\psi\rangle = I|\psi\rangle = \sum_j |b_j\rangle \langle b_j|\psi\rangle$, and regrouping each term $|b_j\rangle (\langle b_j|\psi\rangle)$ as the scalar $\langle b_j|\psi\rangle$ times the vector $|b_j\rangle$ recovers the same formula. This is exactly why $\langle b_j|\psi\rangle$ is called the “component of $|\psi\rangle$ along $|b_j\rangle$ ” – it plays the same role as a dot product picking out coordinates in ordinary vector geometry, just generalized to any orthonormal basis of the state space (not only $\{|0\rangle, |1\rangle\}$).